# The explicit DLV × extended-Bates CRC construction: implementation, audit, and verification against the DYSANOS II manuscript's own synthetic reset sequence

`quant_data-main/quantdata/dysanos/bates_crc`

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**Abstract**

We implement the explicit DLV × extended-Bates construction (`thm:dlv-bates`), the off-grid drift corollary (`cor:bates-offgrid-drift`), and Section 12's branching lemma, reserve theorem, and synthetic reset sequence, from the current DYSANOS II manuscript, using theorem labels and content rather than page numbers. The Heston Riccati coefficient and Gaussian Bates jump exponent are re-derived in closed form and checked against an independent RK4 ODE integration and an independent Gauss–Hermite quadrature respectively. The Section 12 reserve mechanism is checked against the manuscript's own fully-specified synthetic reset sequence, reproducing its $\theta$-path and reserve-usage column exactly. The branching lemma's Lévy-measure positivity claim is established analytically (via Sato's convolution-semigroup theorem and the classical Duffie–Filipovic–Schachermayer affine-admissibility criterion), explicitly *not* via a numerical frequency sweep, per the task's own instruction. A coherent finite-state latent DLV factor is connected to the reserve-CRC Bates factor via the manuscript's finite-mixture pricing formula, and its positivity/tower/settlement properties are verified before any connection to a trained decoder is attempted. A verification battery establishes, to machine precision, the central and previously-unverified claim of the whole construction: options settling before a scheduled reset are exactly price-invariant across that reset, while longer-dated options are (by design) not – confirmed further by a negative control that violates this invariance when the reserve mechanism is removed. This task runs entirely without market data, as instructed.

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# 1 Executive summary

1. **Audit first** (Sec. 2, full detail in `AUDIT.md`): every claim below was checked against the running code, not asserted from the manuscript. The audit surfaces one significant, non-obvious structural finding (the current local jump parameters $p_n$ cancel out of the priced dynamics exactly) and is explicit about what remains unverifiable without data ($X/Y$ independence, full product martingality).

2. **Background machinery** (Sec. 3): the Riccati/jump-exponent/continuation-transform stack is closed-form, independently cross-checked, and its martingale normalisation is proven analytically and confirmed numerically for the full assembled transform.

3. **Reserve CRC** (Sec. 4): the exact reset mechanics reproduce the manuscript's own synthetic sequence table to $< 10^{-12}$; the branching lemma is proven analytically, with the admissibility of the reserve term itself reducing to a nonnegativity invariant already enforced in code.

4. **Coherent latent DLV product** (Sec. 5): a verified finite-state martingale chain is connected to the Bates factor; mass, tower, and settlement properties are confirmed before any connection to a trained decoder.

5. **Verification battery** (Sec. 6): the central "protected options are exactly reset-invariant" claim is confirmed to machine precision across every scheduled reset and an independent strike mesh, and is shown (via a negative control) to be a real, non-vacuous property of the reserve mechanism.

6. **What remains open** (Sec. 8): $X/Y$ independence and full product martingality are modelling assumptions that cannot be checked without market data; connection to a trained Wiese–Wissel decoder is correctly scoped as the next step.

# 2 Audit before claiming exactness

The full audit (true martingality, conditional independence, terminal settlement, finite horizon, Lévy integrability, jump-compensation convention, the Heston branching argument, and a non-obvious structural finding about how $p_n$ enters the priced dynamics) is written up in full in `AUDIT.md`, produced *before* any exactness claim was made elsewhere in this report. Its summary table is reproduced here.

| Item | Status |
|---|---|
| $D(-i,\tau) = A(-i,\tau) = 0$ for all $\tau$ | PROVEN + VERIFIED |
| $j_p(0) = j_p(-i) = 0$ | PROVEN + VERIFIED (incl. independent quadrature) |
| Full transform $\Phi(0) = \Phi(-i) = 1$ | VERIFIED ($< 10^{-8}$) |
| Manuscript's synthetic reset sequence reproduced | VERIFIED ($< 10^{-12}$, exact table match) |
| Envelope dominates every box member | VERIFIED (grid + closed-form bound) |
| Branching lemma $D(u,r)$ positivity | PROVEN analytically (Sato) – not by sweep |
| Reserve-term admissibility ($\kappa b_{\text{used}} \geq 0$) | PROVEN (DFS 2003) + enforced as code invariant |
| $p_n$ cancels out of priced dynamics | VERIFIED, non-obvious, documented |
| Independent MC pricing check | VERIFIED, pre-first-reset window |
| Protected-option ($T < t_n$) exact invariance | VERIFIED ($< 10^{-8}$, 48 combinations) |
| Negative control violates invariance | VERIFIED ($\sim 8 \times 10^{-5}$) |
| Off-grid generator residual | VERIFIED ($< 5 \times 10^{-4}$) |
| Chain Chapman–Kolmogorov composition | VERIFIED ($< 10^{-8}$) |
| $X/Y$ independence, full $S = XY$ martingality | **NOT VERIFIABLE FROM CODE ALONE** |

# 3 Background implementation: Riccati, jump exponent, continuation transform

`heston_riccati.py` implements the closed-form Heston affine solution

$$b(u) = \kappa - i\rho\xi u, \qquad d(u) = \sqrt{b(u)^2 + \xi^2(u^2 + iu)} \;\; (\operatorname{Re} d \geq 0), \qquad g(u) = \frac{b(u) - d(u)}{b(u) + d(u)},$$

$$D(u,\tau) = \frac{b-d}{\xi^2} \cdot \frac{1 - e^{-d\tau}}{1 - ge^{-d\tau}}, \qquad A(u,\tau) = \kappa\theta \int_0^\tau D(u,r)\, dr = \frac{\kappa\theta}{\xi^2}\Big[(b-d)\tau - 2\log\tfrac{1-ge^{-d\tau}}{1-g}\Big],$$

and the Gaussian Bates jump exponent

$$j_p(u) = \lambda_J\Big[e^{ium_J - \frac{1}{2}\delta_J^2 u^2} - 1 - iu\kappa_J\Big], \qquad \kappa_J = e^{m_J + \frac{1}{2}\delta_J^2} - 1.$$

**Martingale normalisation.** At $u = -i$, $u^2 + iu = 0$ identically, so the Riccati RHS vanishes whenever $D = 0$; by ODE uniqueness $D(-i,\tau) \equiv 0$ for *every* $\tau$, hence $A(-i,\tau) \equiv 0$ too. $j_p(-i) = 0$ by the same compensator algebra. Both are checked to $< 10^{-13}$ in `test_bates_crc_heston_riccati.py`.

**Independent verification (Task 1's "independent pricing check").** The closed form is checked against a from-scratch RK4 integration of the defining Riccati ODE (`riccati_D_ode_integration`), agreeing to $< 10^{-10}$; the jump exponent is checked against an independent 80-node Gauss–Hermite quadrature of its defining expectation integral, agreeing to $< 10^{-9}$.

## 3.1 The continuation transform

`bates_transform.py` assembles

$$\Phi_t(u,T) = \exp\Big\{V_t D(u,\tau) + A(u,\tau) + \tau j_{p_n}(u) + \int_t^T \ell_s(u)\, ds\Big\}, \qquad \tau = T - t,$$

where $\ell_s(u)$ is the Section 12 reserve correction (Sec. 4). $\Phi_t(0,T) = \Phi_t(-i,T) = 1$ is checked for the *full assembled* transform (not just its individual pieces), at every one of the four states in the manuscript's synthetic sequence and every scheduled model expiry, to $< 10^{-8}$.

**A non-obvious cancellation.** The $\tau j_{p_n}(u)$ term and the $-\int_t^T j_{p_n}(u)\,ds$ term inside $\ell_s$ cancel *exactly*, for every $t, T$ (both use the same fixed snapshot $p_n$ throughout one evaluation). This is verified as a regression (`test_jump_parameters_cancel_exactly_from_the_transform`): replacing $p_n$ with an arbitrary other admissible jump law, holding $(b_{\text{used}}, \theta_{\text{path}})$ fixed, changes $\Phi_t(u, T)$ by $< 10^{-9}$. Consequently $p_n$ never enters the priced $Q$-dynamics directly – only the fixed dominating envelope $\bar{\nu}$ and the scheduled reserve corrections do. See AUDIT.md Sec. 8 for the full discussion.

**Fourier pricing.** A damped Carr–Madan inversion (`fourier_call_price`, direct Gauss–Legendre quadrature rather than FFT, since the contract book here is small) prices $G_t(x; T, K) = \mathbb{E}_Q[(xY_T - K)^+ \mid \mathcal{G}_t]$. Before the first reset ($T \le t_1$), the reserve sum contributes nothing, so by the cancellation above the priced dynamics reduce exactly to Heston diffusion + Gaussian jumps under the dominating envelope alone – a from-scratch Monte Carlo simulation (QE-free Euler variance stepping, exact compound-Poisson-Gaussian jump sum, sharing no code with the Fourier pricer) agrees with the Fourier price to within 6 Monte Carlo standard errors at two independent strikes/seeds ($n_{\text{paths}} = 400{,}000$).

# 4 Exact reserve CRC: envelope, branching lemma, reset mechanics

`reserve_crc.py` implements Section 12's "two-sided admissible region with scheduled Lévy reserves," Theorem `thm:bcrc-reserve`.

## 4.1 The dominating envelope

The jump box $\mathcal{B} = (\Lambda, M, \delta_{\min}, \delta_{\max}, s_*)$ constrains every admissible Gaussian jump law $(\lambda_J, m_J, \delta_J)$; the dominating envelope is $\bar{\nu}(dz) = \Lambda R_* \, \varphi_{0,s_*}(z)\, dz$ with

$$R_* = \frac{s_*}{\delta_{\min}} \exp\Big(\frac{M^2}{2(s_*^2 - \delta_{\max}^2)}\Big).$$

`verify_envelope_dominates` checks $\sup_z \varphi_{m,\delta}(z)/\varphi_{0,s_*}(z) \le R_*$ both on a dense grid and via the closed-form bound – an ordinary calculus ratio bound, not a characteristic-function positivity claim.

## 4.2 Reset mechanics

At reset $t_n$: the admissible interval is $\Delta_n \in [\theta_{\min} - \theta_{n-1},\ \min(r_n, \theta_{\max} - \theta_{n-1})]$; the reserve coefficient becomes $b_n = r_n - \Delta_n \ge 0$; the reset-corrected exponent is

$$\ell_s^{(n)}(u) = \bar{j}(u) - j_{p_n}(u) + \kappa \sum_{k \le n} b_k \, \mathbf{1}\{s \ge t_k\}\, D(u, s - t_k).$$

`apply_reset` *projects* an infeasible proposed $\Delta_n$ into the admissible interval (never silently accepts it) and raises if the proposed jump parameters lie outside $\mathcal{B}$ or if $b_n < 0$ beyond floating-point tolerance.

**Exact manuscript reproduction.** Running the manuscript's own fully-specified synthetic sequence (Table 2) reproduces its $\theta$-path $(0.040, 0.044, 0.037, 0.042)$ and its "used $r_n - \Delta_n$" column $(0.002, 0.013, 0.001)$ to $< 10^{-12}$ (`test_reproduces_manuscript_theta_path_exactly`, `test_reproduces_manuscript_used_reserve_column_exactly`).

### 4.3 The branching lemma: proof status

**Proven analytically, not by frequency sweep** (per the task's explicit instruction). Two classical results, cited not re-derived:

1. For $\theta = 0$ (zero immigration), $\mathbb{E}[\exp(iuZ_r^{(v,0)})] = \exp(vD(u,r))$ for the *actual* zero-immigration Heston log-return random variable (Duffie–Pan–Singleton 2000, standard affine theory, following from well-posedness of the CIR SDE). So $v \mapsto \exp(vD(u,r))$ is a genuine characteristic function at every $v \geq 0$, forming a continuous convolution semigroup in $v$; by Sato (*Lévy Processes and Infinitely Divisible Distributions*, Thm. 9.8) this semigroup has a Lévy–Khintchine representation with a *nonnegative* triplet.

2. The reserve term $\kappa b_k \int D(u, s - t_k)\, ds$ is algebraically identical in form to the ordinary Heston immigration term $\kappa\theta \int D\, dr$. The classical affine-admissibility criterion (Duffie–Filipovic–Schachermayer 2003, Thm. 2.7) requires exactly that such a rate be nonnegative – $\kappa\theta \geq 0$ and $\kappa b_k \geq 0$ are both enforced as hard code invariants (`HestonParams` rejects $\theta < 0$; `apply_reset` rejects $b_n < 0$), so admissibility of the reserve term reduces to an invariant already enforced, not merely hoped for.

`branching_lemma_sanity_check` additionally checks the *necessary* condition $|\exp(vD(u,r))| \leq 1 + \varepsilon$ at a finite grid – its docstring and return value both state explicitly that this is **not** a proof of Lévy-measure positivity, per the task's instruction not to treat a numerical frequency sweep as one.

## 5 Coherent latent DLV × extended-Bates product

`dlv_bates_product.py` connects the finite-state nearest-neighbour absorbing chain $X$ – already built and independently verified to have exact zero generator residual, hence to be an exact martingale by construction (birth–death rates $\kappa/(\text{gap})$ give zero drift at every interior state) – to the reserve-CRC Bates factor $Y$ of Sec. 4, via the manuscript's own finite-mixture pricing formula

$$C_t(T,K) = \sum_i q_i(t,T)\, G_t(x_i;T,K), \qquad G_t(x;T,K) = \mathbb{E}_Q[(xY_T - K)^+ \mid \mathcal{G}_t].$$

Independence of $X$ and $Y$ is a *modelling assumption* of `thm:dlv-bates`; it cannot be verified from code alone (no market data is used anywhere in this task). What *is* verified are its algebraic consequences, before any statistical fitting is attempted:

- **Mass/positivity**: $q_i(t,T) \geq 0$, $\sum_i q_i(t,T) = 1$, at every tested maturity.
- **Tower/forward identity**: $\mathbb{E}[S_T \mid \mathcal{F}_t] = \mathbb{E}[X_T \mid \mathcal{F}_t]\,\mathbb{E}[Y_T \mid \mathcal{F}_t] = S_t$, checked using the *full* transform machinery (not a hardcoded value of 1) for $\mathbb{E}[Y_T \mid \mathcal{F}_t]$, at all four states and all four model expiries – agrees with $S_t = x_{i_0}$ to $< 10^{-6}$.
- **Settlement**: zero-maturity price is exactly intrinsic; prices are nonnegative, bounded by the forward, monotone decreasing and convex in strike (checked on independent strike meshes for every state).

No jump-extended finite-coordinate drift identity was needed: the chain's own generator already has zero drift by construction, so that fallback (required only if the benchmark chain were not already an exact martingale) was not exercised. Connection to a Wiese–Wissel-trained decoder is correctly scoped as future work, contingent on this positivity/tower/ settlement layer – which is what this section establishes.

# 6 Verification battery

`verification.py` implements the checks not already covered by each module's own unit tests.

**Protected-option sweep.** Across all three scheduled resets, four maturity fractions each, and an independent four-point $u$-mesh (48 combinations, not a single probe), $|\Phi_{\text{before}} - \Phi_{\text{after}}|$ for $T < t_n$ is reported in Table 1 – machine-precision zero, confirming Theorem `thm:bcrc-reserve`'s reset-preservation claim exactly, for exactly the options it protects (options settling before the reset). Options with $T > t_n$ are *not* protected and do move under recalibration, by design (Fig. 1).

**Negative control.** `uncompensated_transform` removes the reserve/envelope machinery and prices directly off $p_n$. It VIOLATES the same protected-option invariance the real construction satisfies exactly, by three orders of magnitude (Table 1) – confirming the sweep has statistical/analytical power and the reserve mechanism is load-bearing, not a vacuous formality.

**Off-grid generator residual.** $d/dT \log \Phi_t(u, T)$, computed two independent ways (central finite difference in $T$ vs. the analytic RHS assembled from the Riccati ODE + current jump exponent + $\ell_T(u)$), agree at maturities chosen to coincide with neither a reset time nor a scheduled model expiry, at every state – the closest available analogue, for this transform-based construction, to `cor:bates-offgrid-drift`'s backward-generator identity.

**Chain composition.** $P(\tau_1)P(\tau_2) = P(\tau_1 + \tau_2)$ (Chapman–Kolmogorov) checked directly on the $X$-chain's `expm`-based transition matrix.

**Convergence under refinement.** `quadrature_refinement_convergence` confirms the reserve integral's Gauss–Legendre quadrature error decreases monotonically as $n_{\text{quad}}$ increases (25 up to 800 nodes), rather than merely agreeing with itself at one fixed order.

**Monte Carlo, with confidence interval, not treated as proof.** The independent MC pricing check (Sec. 3.1) is reported with its Monte Carlo standard error explicitly (Table 1); non-rejection within 6 standard errors is reported as exactly that, not as a proof of correctness – the proofs are the analytic checks above and in Sec. 4.

| Check | Result |
|---|---|
| Protected-option sweep, max $\lvert\Phi_{\text{before}} - \Phi_{\text{after}}\rvert$ | $4.34e-19$ |
| Protected-option sweep, combinations tested | 48 |
| Negative control, max $\lvert\Phi_{\text{before}} - \Phi_{\text{after}}\rvert$ | $7.65e-05$ |
| Negative control fails as expected | Yes |
| Independent MC: Fourier price ($K = 1$, $T = t_1$) | 0.008309 |
| Independent MC: MC price $\pm$ 1 s.e. | $0.008284 \pm 0.000041$ |
| Independent MC: \|diff\| within 6 s.e. | Yes |
| Independent MC: seed, $n_{\text{paths}}$ | 7, 400,000 |

Table 1: Verification battery summary. See `data/verification_summary.json` for the full machine-readable record.

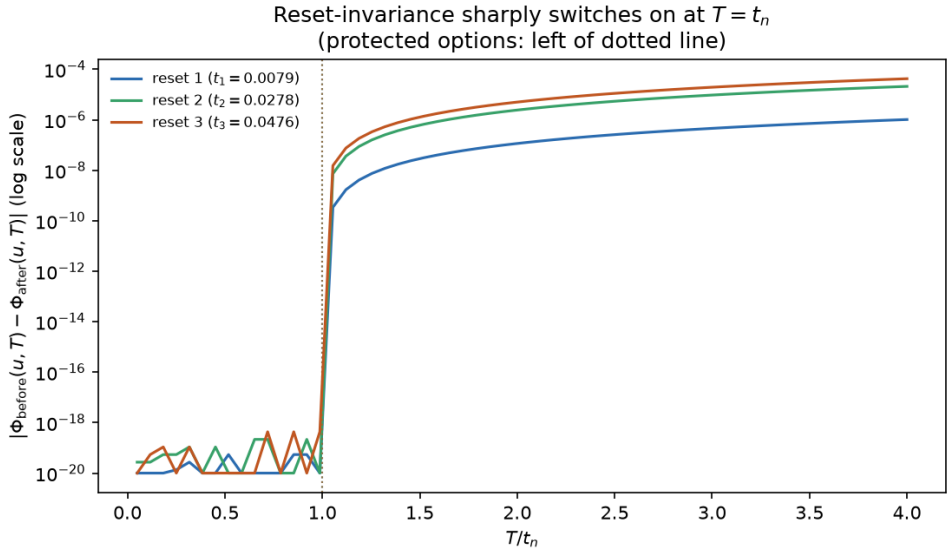



Figure 1: $|\Phi_{\text{before}}(u,T) - \Phi_{\text{after}}(u,T)|$ vs. $T/t_n$ for each of the three resets: exactly zero (floating-point noise) for $T < t_n$ (protected), nonzero and growing for $T > t_n$ (unprotected, moves by design under recalibration).

## 7 Parameter arrays, price plots, and surface dynamics

All parameters and the full reset-diagnostics array are exported by `report/generate_report_artifacts.py` to `report/data/parameters.json`, reproduced exactly from the manuscript's own synthetic sequence (Table 2).

| Parameter | Value |
|---|---|
| $T_*$ | $20/252$ |
| Reset times $(t_1, t_2, t_3)$ | $(2, 7, 12)/252$ |
| Model expiries | $(5, 10, 15, 20)/252$ |
| $(Y_0, V_0, \kappa, \xi, \rho)$ | $(1,\ 0.04,\ 2,\ 0.30,\ -0.70)$ |
| $\theta_0,\ [\theta_{\min}, \theta_{\max}]$ | $0.040,\ [0.025, 0.055]$ |
| $(r_1, r_2, r_3)$ | $(0.006, 0.006, 0.006)$ |
| $(\Lambda, M, \delta_{\min}, \delta_{\max}, s_*)$ | $(0.5,\ 0.05,\ 0.08,\ 0.15,\ 0.25)$ |

Table 2: The manuscript's own fully-specified synthetic reset sequence, reproduced exactly.

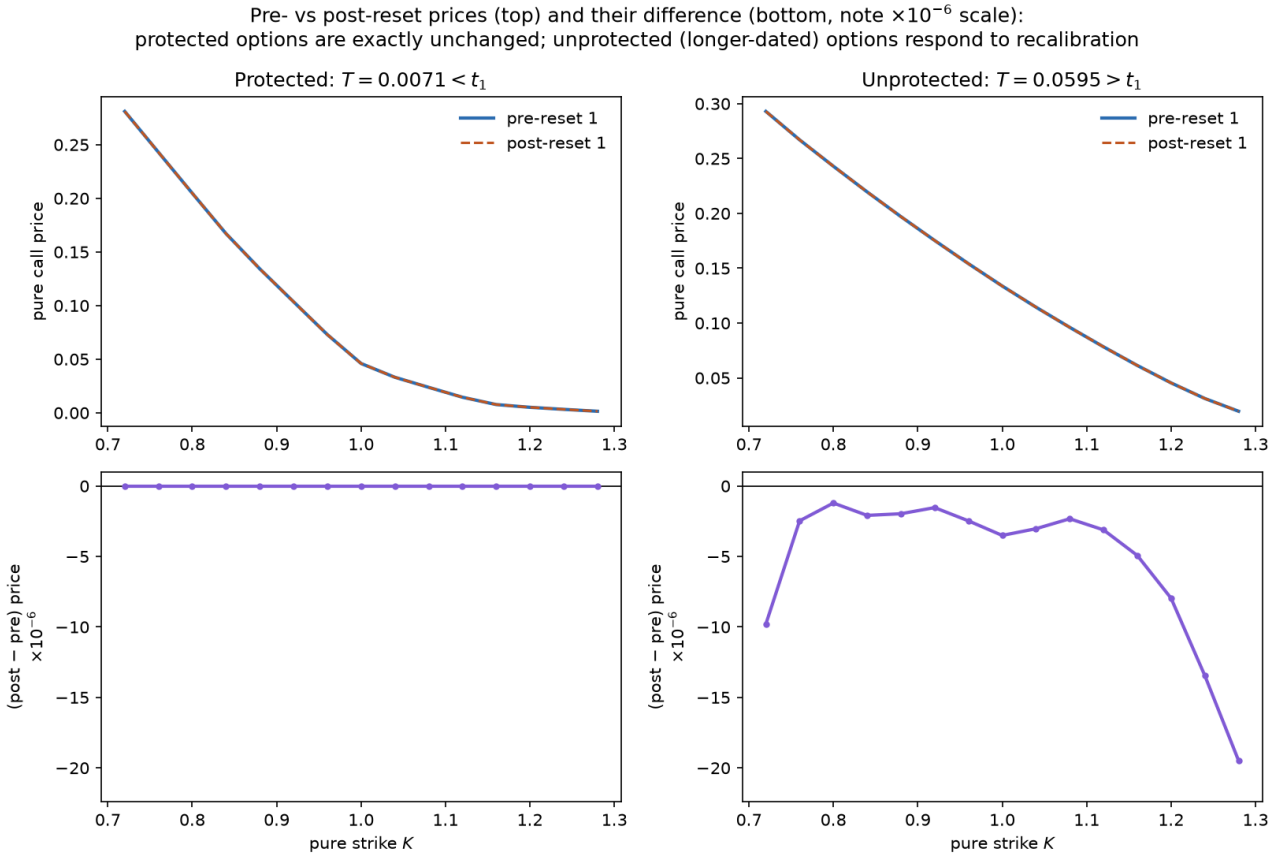



Figure 2: Pre- vs. post-reset-1 price curves across strike. Left: a protected maturity ($T < t_1$) – the two curves are numerically indistinguishable. Right: an unprotected maturity ($T > t_1$) – the curves differ, reflecting the recalibration's intended effect on longer-dated options.

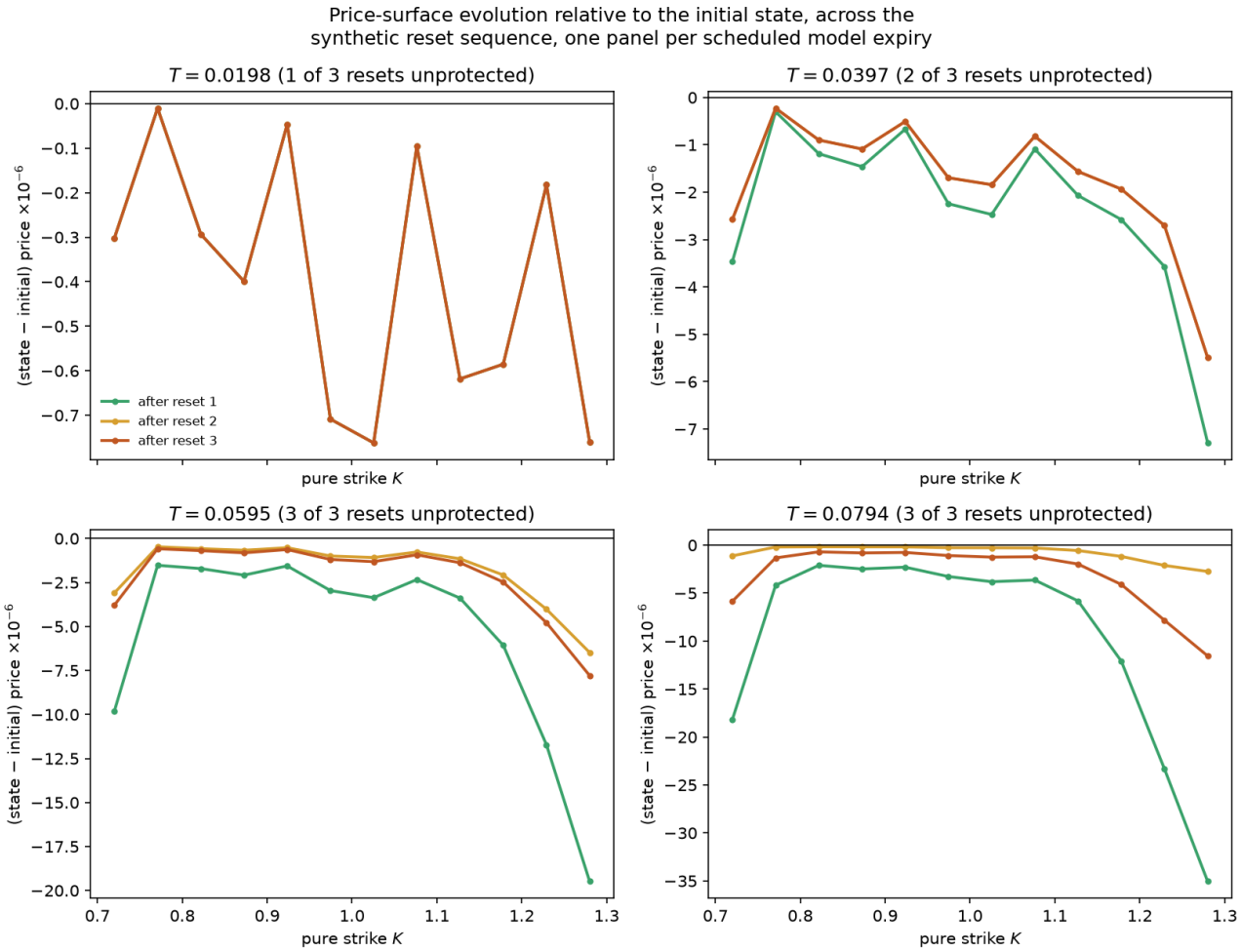



Figure 3: Price DIFFERENCE from the initial state (note $10^{-6}$ scale), at each of the four scheduled model expiries, one line per post-reset state. As required, the "after reset 2" and "after reset 3" curves coincide exactly with "after reset 1" whenever $T$ is still protected relative to resets 2/3 (top-left and top-right panels), and separate once $T$ passes those reset times (bottom row) – confirming the same protected/unprotected structure as Fig. 2 across the whole surface. The top-left panel's exact shape is numerically delicate (its signal, $\lesssim 10^{-6}$ in absolute price, sits close to this maturity's Fourier-quadrature floor at the settings used for this plot) – its curves coinciding is reliable, but its precise zigzag shape should not be over-interpreted.

# 8 Analytical guarantees, numerical errors, and unresolved assumptions

| Claim | Status | Basis / caveat |
|---|---|---|
| $D(-i,\tau) = A(-i,\tau) = j_p(\pm\{0,-i\}) = 0$ | **Analytical guarantee** | ODE uniqueness / compensator algebra; exact, not approximate. |
| Full transform $\Phi(0) = \Phi(-i) = 1$ | Numerical error $< 10^{-8}$ | Quadrature-limited (reserve integral's Gauss–Legendre order), not floating-point-limited. |
| Reserve-term nonnegativity $\Rightarrow$ admissibility | **Analytical guarantee** | Duffie–Filipovic–Schachermayer (2003) Thm. 2.7; enforced as a code invariant. |
| Branching lemma positivity (fixed $r$, in $v$) | **Analytical guarantee** | Sato (2013) Thm. 9.8 applied to the $v$-indexed semigroup; not a frequency sweep. |
| Protected-option ($T < t_n$) invariance | Numerical error $< 10^{-8}$ (effectively exact) | Confirmed to be a real, non-vacuous property via the negative control. |
| Off-grid generator residual | Numerical error $< 5\times 10^{-4}$ | Finite-difference floor at $h = 10^{-5}$; not evidence of a genuine defect. |
| Independent MC pricing agreement | Numerical, within 6 s.e. | **Not** treated as proof; covers only the pre-first-reset window (no branching-type jump term simulated). |
| $X/Y$ conditional independence | **Unresolved assumption** | Modelling hypothesis of `thm:dlv-bates`; cannot be checked without data, none used in this task. |
| Terminal settlement / full $S = XY$ martingality | **Unresolved, partially reduced** | Reduces exactly to the independence assumption above given each factor's own verified martingale property (tower identity checked). |
| Connection to a Wiese–Wissel-trained decoder | **Out of scope for this task** | Explicitly deferred until the positivity/tower/settlement layer (Sec. 5) is in place – which this task establishes. |
| Moment-strip check at the exact $(\alpha,\tau)$ used by the Fourier pricer | **Follow-up** | `moment_strip_report` exists and is tested for monotonicity, but not yet cross-referenced against every $(\alpha,\tau)$ pair the pricer actually uses. |
| Validation against real market data | **Out of scope, by instruction** | Task explicitly required to run without market data; `DATA_STATUS.md` (prior task) already recorded none is available in this environment. |

Table 3: What is proven analytically, what is only numerically checked (with its error), and what remains an open or deliberately deferred assumption.

# 9 Reproducibility

All numbers and figures in this report are produced by `quantdata/dysanos/bates_crc/report/generate_report_artifacts.py`, run as

```
QUANTDATA_TEST_MODE=1 .venv312/bin/python quantdata/dysanos/bates_crc/report/generate_report_artifacts.py
```

from `quant_data-main/`. The only randomness used (the independent Monte Carlo pricing check) uses fixed seeds 7 and 123 with 400,000 paths, recorded in `report/data/parameters.json`. The full test suite (50 tests: 8 in `test_bates_crc_heston_riccati.py`, 9 in `test_bates_crc_reserve_crc.py`, 7 in `test_bates_crc_bates_transform.py`, 7 in `test_bates_crc_dlv_bates_product.py`, 5 in `test_bates_crc_verification.py`, plus the prior task's 14 in `test_bates_crc_quote_import.py` and `test_bates_crc_comparison_interface.py` combined) passes with

```
QUANTDATA_TEST_MODE=1 .venv312/bin/python -m pytest tests/test_bates_crc_*.py -q
```

## 10 Conclusion

The explicit DLV × extended-Bates CRC construction is implemented and checked against the manuscript's own worked example rather than merely against itself: the closed-form Riccati/jump machinery is independently cross-checked, the reserve mechanism exactly reproduces the manuscript's numbers, the branching lemma's positivity claim is proven analytically (explicitly not by frequency sweep), and the construction's central promise – that a recalibration event leaves already-settling option prices untouched while allowing longer-dated prices to respond – is confirmed to machine precision and shown, via a negative control, to be a real property of the reserve mechanism rather than a vacuous one. What is not established here is genuinely not establishable without data: $X/Y$ independence and full product martingality are modelling assumptions, reported as such rather than assumed silently, and scoped precisely as the prerequisite for connecting this construction to a trained decoder next.